

Income and Deductions

When income is earned through employment, various **deductions** are taken from it before it is paid into an employee's bank account. Each time an employee is paid, their employer should issue them with a **payslip** detailed how both their income and deductions were calculated, as well stating clearly how much **net pay** is being paid into their bank account. For example:

Example Company Ltd
10 Example Street, London, AB1 2CD

PAYSIP

Employer	Example Company Ltd	Pay period	1 August–31 August 2026
Payment date	31 August 2026	Pay frequency	Monthly
PAYE reference	123/AB456	Payroll number	EMP-0042

Employee Details			
Employee name	Alex Example	NI number	QQ 12 34 56 C
Job title	Classroom Teacher	Tax code	1257L
Department	Education	NI category	A

Payments	Amount	Deductions	Amount
Basic salary	£ 3,250.00	PAYE income tax	£ 456.20
Overtime	£ 125.00	National Insurance	£ 178.64
Commission	£ 75.00	Workplace pension	£ 172.50
Bonus	£ 0.00	Student loan	£ 45.00
Other payments	£ 0.00	Other deductions	£ 0.00
Gross pay	£ 3,450.00	Total deductions	£ 852.34

NET PAY	£ 2,597.66
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This chapter will cover various calculations related to income and deductions.

3.1 Gross Pay

The income an employee earns from their employment **before any deductions have been made** is called *gross pay*. This can be made up of:

- **Salary:** A fixed amount of money agreed between an employer and an employee, usually paid at regular intervals such as weekly or monthly.
- **Hourly Pay:** Money earned according to the number of hours worked. The total is calculated by multiplying the employee's hourly rate by the number of hours worked.
- **Overtime Pay:** Additional money earned for working beyond the employee's normal contracted hours. Overtime may be paid at the standard hourly rate or at a higher rate, such as time-and-a-half.
- **Bonus:** An additional payment awarded on top of normal wages or salary. Bonuses may be linked to individual performance, company results, attendance, or other achievements.
- **Commission:** A payment based on the value or number of sales made by an employee. Commission may be paid in addition to a basic salary or hourly wage.

The amount of pay a job advertises is typically the **annual salary**, which is the amount paid across one full year of work. This is typically paid on a monthly basis, although some jobs instead pay earn week or fortnight (two weeks). When converting between intervals of time, the typical calculations required consider:

- There are **12 months in a year**.
- There are **52 weeks in a year**.

Whilst there are *approximately* four weeks in a year, this is not accurate enough to be used when calculating pay.

Example 1

An employee earns an annual salary of £43,000 as well as 8% commission on the total volume of sales they made and a bonus of £100 in any month during which their sales exceed £5,000.

In August they made sales worth a total of £6,200. Calculate their gross pay for August.

Your working should be suitably organised and labelled, making it clear and easy to follow (and award marks for).

When working with hours as a decimal, remember that:

- 15 minutes is 0.25 hours.
- 30 minutes is 0.5 hours.
- 45 minutes is 0.75 hours.

Example 2

An employee is paid at an hourly rate of £14.50. In September, they worked from 9am to 5pm for a total of 21 working days with an unpaid lunch break of 45 minutes during each of those.

Calculate their gross pay for September.

Overtime pay is often either *double-pay* (for which the hourly rate is multiplied by 2), or *time-and-a-half* (for which the hourly rate is multiplied by 1.5).

Example 3

An employee works 35 hours each week and is paid £16.00 per hour. Any hours worked beyond 35 hours are paid at time-and-a-half.

In one week, the employee worked a total of 43 hours. Calculate their gross pay for that week.

Example 4

An employee is paid £15.00 per hour for the first 40 hours worked each week. Any hours worked beyond 40 hours are paid at time-and-a-half. The employee also receives commission equal to 5% of the value of their weekly sales.

In one week, they worked 46 hours and made sales worth £3,600. Calculate their gross pay for that week.

3.2 Pension Contributions

When people in the UK reach their *state pension age*, they become eligible to start receiving a *state pension* provided they have met certain conditions, usually involving having worked for a minimum number of years. For many adults, the current projected state pension age is 68 years of age.

Many people during their adult lives additionally pay money into a **private pension**, which they will receive in addition to their state pension. It can also be possible to begin *taking a pension* at an earlier age than the state pension age, allowing earlier retirement. Whilst anyone can pay into a private pension, the most popular are typically *workplace pensions*, since:

- For each *contribution* of money you (the employee) pay into your workplace pension, your employer will typically also make an *employers' contribution* into your *pension pot*.
- Money paid directly into a pension from your income will not have tax subtracted from it.

Some *pension schemes* allow flexibility in what percentage of income is paid as a *pension contribution*, whilst others may specify the percentage required. Whilst it is typically possible to *opt-out* of a workplace pension scheme, in the long run it is **strongly recommended** to take advantage of their benefits to enjoy a more comfortable retirement.

Example 1

Sophie earns a monthly salary of £2800 and she pays 3.5% of her earnings into her pension fund. Her employer also contributes 7.2% of the same earnings into her pension fund. Calculate the total amount paid into Sophie's pension fund **annually**.

Once an individual's *pension contributions* are subtracted from their *gross pay*, the amount remaining is referred to as **taxable pay**, since that is what income tax is calculated based on. Note that *employers' contributions* are not part of this calculation.

Example 2

Bob earns an annual salary of £32,400 and he pays 2.8% of her earnings into his pension fund. His employer also contributes 5.4% of the same earnings into his pension fund. Calculate Bob's **monthly** taxable pay.

3.3 National Insurance

National Insurance is a kind of tax deducted from an employee's gross pay. The Higher Applications of Maths exam shares a **Data Booklet** ahead of the final exam containing up-to-date details on how *National Insurance contributions* are calculated from gross pay, as well as other information. For example, from the 2026 data booklet:

Data Booklet: National Insurance contributions

You begin paying National Insurance once you earn more than £1048 a month (this is the amount for the 2025/26 tax year).

For payslips dated between 6 April 2025 and 5 April 2026, you pay 8% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

National Insurance is calculated on a person's salary before deductions such as pension contributions.

To calculate NI contributions, you need to calculate how much of the gross pay falls into each of the *NI bands*. Since the data given relates to **monthly** pay, it is recommended to start by considering one *month*.

Example 1

Betty earns an annual salary of £51,888 and she pays £3,750 into pension fund each year. Calculate her monthly National Insurance contributions.

Example 2

Sal earns a monthly salary of £5134. Calculate their total annual National Insurance contributions.

3.4 Income Tax

Like National Insurance, income tax is calculated in *tax bands*, and the percentage taken from each band is given in the Data Booklet. Unlike National Insurance, the values given are based on **annual salaries**, and income tax is calculated based on *taxable pay*, which is gross pay **minus any pension contributions**.

Data Booklet: Scottish tax bands

Band	Taxable Income	Scottish tax rate
Personal Allowance	Up to £12,570	0%
Starter rate	£12,570 – £15,397	19%
Basic rate	£15,397 – £27,491	20%
Intermediate rate	£27,491 – £43,662	21%
Higher rate	£43,662 – £75,000	42%
Advanced rate	£75,000 – £125,140	45%
Top rate	Over £125,140	48%

To calculate income tax, you need to calculate how much of the taxable pay falls into each of the *tax bands*. Since the data given relates to **annual** pay, it is recommended to start by considering a *year*.

Example 1

Clara earns an annual salary of £32,000 and she pays £2,768 into her workplace pension scheme each year. She pays £1,554.40 per year in National Insurance contributions. Calculate the annual tax she should pay.

Example 2

An office worker earns an annual salary of £62,000 and does not pay any money into a pension fund. They pay £275.88 each month in NI contributions. Calculate their monthly income tax payments.

3.5 Net Pay

Net pay is the money remaining to be paid into your bank account after all deductions have been made from your *gross pay*. The main deductions are typically pension contributions, National Insurance contributions and income tax, although some people may have other deductions included, such as:

- **Student Loan Repayments:** Repayments deducted from pay when an employee has an outstanding student loan and earns above the relevant repayment threshold.
- **Trade Union Subscription:** Membership fees paid to a trade union. Some employers allow these subscriptions to be deducted directly from an employee's pay.
- **Attachment of Earnings:** A deduction that an employer is legally required to make under an official order. This could be used to collect certain unpaid debts, council tax arrears or maintenance payments.

Example 1

Jamie earns an annual salary of £38,000. They pay annual National Insurance contributions of £2,034 and make annual workplace pension contributions of £1,900. Calculate Jamie's annual net pay.

Example 2

Priya earns £3,200 per month. Her monthly Income Tax has already been calculated as £438.60. She also contributes 5% of her gross monthly pay to her workplace pension. Calculate Priya's monthly net pay.

Chapter 3 Review

Use this checklist when revising for any assessment/exam. Please ask for help with anything you don't understand.

	Skills & Knowledge	Practice Questions			
3.1 Gross Pay	I can convert between weekly/monthly/annual salaries. I can calculate hourly pay and overtime pay. I can calculate commission.				
3.2 Pension Contributions	I know that pension contributions are based on gross pay. I can calculate pension contributions.				
3.3 National Insurance	I know than National Insurance is calculated on gross pay. I can calculate National Insurance contributions.				
3.4 Income Tax	I know that income tax is calculate on taxable pay. I can calculate income tax payments.				
3.5 Net Pay	I can can calculate net pay including all deductions.				

Notes